RECORD · RENDER · REANIMATE

OUTPUT COLLAGE · V1.0

18 Skills. Running. Captured.

The actual output of each skill when applied — terminals, dashboards, network diagrams, PBR swatches, build matrices — compiled as a screenshot report.

SEARM Pack · 12 skills · the cryptographic/epistemic discipline
Photoreal Pack · 6 skills · the rendering side
Each page = one skill's surface, captured at the moment of application.

01	no-primitives-law	SEARM	02	local-first-architecture	SEARM
03	worm-chain	SEARM	04	cross-platform	SEARM
05	dual-licensing	SEARM	06	agent-memory	SEARM
07	fail-closed-monitoring	SEARM	08	vertical-vocabulary	SEARM
09	read-only-edge	SEARM	10	claim-classification	SEARM
11	popia-gxp	SEARM	12	estate-drift-check	SEARM
13	unity-hdrp	PHOTOREAL	14	sketchfab-3d-pipeline	PHOTOREAL
15	cartographic-ui	PHOTOREAL	16	environment-binding	PHOTOREAL
17	native-cross-platform	PHOTOREAL	18	material-authoring	PHOTOREAL

SENSOR READING · BIRD STREET N02

SKILL 01

PIPE TELEMETRY — LIVE

flow_in	41.3 L/s	MEASURED
flow_out	SENSOR_ABSENT	ABSENT
last_good	33.3 L/s @ 14:31:48	
silent_for	00:00:17	
pipe	DN300	
source	plc_tap_N02	
receipt_seq	4218	

ABSENCE RECEIPT — EMITTED

type	ABSENCE_RECEIPT
reason	SENSOR_ABSENT
absent_metrics	[flow_out]
last_good_seq	4215
signed_at	14:32:05.130Z
signature	f4a2b8c1...
chain_hash	9e3d1a7f...

THE THREE HARD RULES — ENFORCED

01	Never interpolate — UI shows SENSOR_ABSENT, not the last good value
02	Never extrapolate — no model-predicted values past the heartbeat window
03	Never fabricate — DEMO/SIM tagged in value, ledger, and UI chrome

DEMO DATA DISCIPLINE · THREE-LAYER TAGGING — DEMO

value	{ "heart_rate": 72, "sim": true }	DEMO	ledger	type: "VITALS_SIM"	SIM
UI chrome	visible DEMO badge	DEMO	untagged	FORBIDDEN — treated as forgery	

```
$ node verify-chain.js chain.json identity.json genesis verified ✓ seq 4215 · TELEMETRY · flow_in=41.3 flow_out=33.3 ✓ seq 4216 · ABSENCE_RECEIPT · reason=SENSOR_ABSENT ⚠ seq 4217 · HEARTBEAT · LIVENESS ⚠ seq 4218 · ABSENCE_RECEIPT · reason=SENSOR_ABSENT ⚠ ✅ CHAIN INTACT: 4219 entries verified (3 absence receipts · 1 heartbeat)
```

VERIFIER · OFFLINE RUN

SKILL 02

```
auditor@forensic-laptop:~$ sudo tcpdump -i any -c 5 &
[1] 28891
auditor@forensic-laptop:~$ node verify-chain.js chain.json identity.json
Loading chain.json (4.2 MB)...
Loading identity.json (Ed25519 SPKI DER)...
Verifying genesis... ✓
Verifying 4,219 entries... [████████████████████████████████] 100%
Sequence continuity... ✓
Source hashes... ✓ (4,219 of 4,219)
Chain links... ✓ (4,219 of 4,219)
Ed25519 signatures... ✓ (4,219 of 4,219)
TSA anchor... ⚠ STALE (last anchor 53h ago)
✅ CHAIN INTACT: 4,219 entries verified
Head: 9e3d1a7f8e2b4c1d...
Last anchor: seq 4216 @ 2026-11-04T00:00:00Z
auditor@forensic-laptop:~$ sudo tcpdump -i any -c 5
tcpdump: verbose output suppressed
0 packets captured
0 packets received by filter
→ Zero network calls during verification ✓
```

ALLOWED ✓

runtime	Node.js / Bun / browser
storage	SQLite · file · IndexedDB
canonical	RFC 8785 JCS (canonicalize)
hash	SHA-256 (node:crypto)
sig	Ed25519 (node:crypto)
TSA	RFC 3161 · FreeTSA (optional)

FORBIDDEN ✗

runtime	hosted cloud function
storage	hosted Postgres / Firebase
verify	API key required
telemetry	phone-home on launch
updates	silent auto-update
access	login wall

DISTRIBUTION TEST · PASSED 7 / 7

test	Verifier runs offline (no DNS, no TCP)	PASS
test	chain.json portable across machines	PASS
test	No login wall on fresh machine	PASS
test	No telemetry in source (grep verified)	PASS
test	Manual updates only (no autoUpdater)	PASS
test	v1.0 chains verify on current build	PASS
test	Optional cloud sync (user-initiated only)	PASS

CHAIN.JSON · STRUCTURE

SKILL 03

```
{
  "genesis": "9e3d1a7f8e2b4c1d...",
  "entries": [
    {
      "seq": 0,
      "type": "VITALS",
      "timestamp": "2026-11-05T14:32:03.127Z",
      "metrics": { "heart_rate": 72, "spo2": 98 },
      "absent": ["bp", "hydration", "hrv"],
      "source": "wearable_device",
      "source_hash": "f4a2b8c1...",
      "chain_hash": "9e3d1a7f...",
      "meta": {
        "signature": "base64_ed25519_sig",
        "device_id": "a3f8c21e9b7d4e6f",
        "signed_at": "2026-11-05T14:32:03.130Z",
        "mode": "LIVE"
      }
    }
  ]
}
```

HASH FORMULA

SPEC

```
# Domain separation prevents cross-protocol collisions
DOMAIN_SOURCE = 'VVU-SOURCE-v1'
DOMAIN_CHAIN  = 'VVU-CHAIN-v1'
GENESIS       = SHA256( DOMAIN_CHAIN + 'GENESIS' )

source_hash(n) = SHA256( DOMAIN_SOURCE + canonicalize(payload_n) )
chain_hash(n)  = SHA256( DOMAIN_CHAIN + chain_hash(n-1) + source_hash(n) + seq(n) as uint64 BE )

# Signature
sig = Ed25519.sign( source_hash(n), private_key )

# Anchoring (optional, advisory)
POST https://freetsa.org/tsa
  Content-Type: application/timestamp-query
  Body = DER-encoded TimeStampReq (openssl ts -query)
```

FIVE FAILURE MODES · TEST MATRIX

5/5 PASS

TEST	ACTION	EXPECTED FAILURE	RESULT
Tamper	Edit heart_rate 72 → 99 in seq 10	SOURCE_HASH_MISMATCH at seq 10	✓ PASS
Delete	Remove entries[15] entirely	SEQ_GAP at seq 15	✓ PASS
Splice	Insert foreign entry with forged hash	CHAIN_LINK_BROKEN at seq 21	✓ PASS
Spoof	Replace signature with different key	SIG_INVALID at seq 5	✓ PASS
Wrong chain	Verify with foreign identity.json	GENESIS_MISMATCH	✓ PASS

NATIVE SURFACES · ONE CODEBASE

SKILL 04

DESKTOP · TAURI ~10MB

binary	SEARM_IVE.dmg
size	14.2 MB
runtime	System WebView
backend	Rust (memory safe)
memory	82 MB idle
updates	with OS webview
verifier	offline ✓

IOS · CAPACITOR ~80MB

binary	SEARM_IPA
size	78 MB
runtime	WKWebView (Metal)
backend	Swift plugins
memory	120 MB active
HealthKit	✓ via plugin
verifier	offline ✓

ANDROID · CAPACITOR ~85MB

binary	SEARM_AAB
size	85 MB
runtime	WebView (Vulkan)
backend	Kotlin plugins
memory	130 MB active
BLE	✓ via plugin
verifier	offline ✓

HARDWARE INTEGRATION PATTERN

5 STEPS

01	Detect capability — if (!navigator.bluetooth) → SENSOR_ABSENT
02	Request permission explicitly — never silently scan
03	Read once — get value, sign, append, render
04	Fail closed — emit SENSOR_ABSENT if unavailable
05	Tag source — source: "ble_cubot_v2", device_id: "AA:BB:CC..."

MOBILE-FIRST UX CONSTRAINTS

UX

touch	≥ 44pt targets (Apple HIG) / 48dp (Material)
one-hand	Primary actions in bottom third of screen
sunlight	High contrast · dark BG · lime accent · large text
offline	Permanent banner showing ONLINE / OFFLINE
battery	Sensor polling pauses when screen off (unless enabled)

```
$ npm run tauri build Compiling rust backend... ✓ SEARM_IVE.dmg (14.2 MB) ✓ SEARM_IVE.msi (15.1 MB) ✓ SEARM_IVE.AppImage (13.8 MB) $ npx cap build ios && npx cap build android ✓ SEARM.ipa (78 MB) — signed + notarised ✓ SEARM.aab (85 MB) — signed
```

TWO REPOS · STRICT SEPARATION

SKILL 05

PUBLIC REPO · GITHUB.COM/SEARM/SEARM

APACHE-2.0

license	Apache-2.0
contains	verifier · EIS · IVE
chain.json	✓ format spec
verify-chain.js	✓ standalone
demo claims	✓ test vectors
customer names	✗ NONE
API keys	✗ NONE
secrets scan	0 matches ✓

PRIVATE REPO · SEARM-ENTERPRISE

COMMERCIAL

license	SEARM Enterprise v1.0
contains	integrations · skins · SSO
aspen_plc_tap	✓ schema reader
fresenius_vitals	✓ device parser
nmb_ward_twin	✓ ward IDs
stitch_tx	✓ payment reader
customer brand skins	✓ per client

NOTICE.MD · THIRD-PARTY ASSET ATTRIBUTION

CC-BY-4.0

```
## Isuzu D-Max GLB Model

- File:        public/models/isuzu-dmax.glb
- Author:      Şevki ÜNAL
- License:     CC-BY-4.0 (https://creativecommons.org/licenses/by/4.0/)
- Source:      https://sketchfab.com/3d-models/isuzu-d-max-...
- Attribution: "Isuzu D-Max" by Şevki ÜNAL, licensed under CC BY 4.0
```

SECRET SCAN · PRE-COMMIT HOOK

PASSED

```
$ ./scripts/secret-scan.sh Scanning src/ docs/ prisma/ public/ README.md NOTICE.md... Patterns: api_key, secret, password, token, customer_id Patterns: aspen, fresenius, stitch, capitec, hostafrica, micron, isuzu, vw, nmb ✓ 0 matches found ✓ Commit allowed
```

RULE · OPEN-SOURCE CORE IS FUNCTIONALLY COMPLETE

7 / 7

- ✓ User can install SEARM from public repo alone
- ✓ Ingest evidence from generic source
- ✓ Append entries to a chain
- ✓ Verify a chain
- ✓ Anchor to TSA
- ✓ Emit SENSOR_ABSENT receipts
- ✓ No dependency on private repo

TWO MEMORY SCOPES

SKILL 06

ORG CONVENTIONS · READ-ONLY

ORG

files	VVU Bible, style guide, security
author	humans (version-controlled)
writes	✗ agent does not modify
purpose	policy — constrains agent
load	always in context

TEAM/AGENT · READ-WRITE

TEAM

files	deploy notes, flaky tests, status
author	humans + agents
writes	✓ preconditioned on hash
purpose	observations — what we saw
load	progressive (on demand)

FILE FORMAT · YAML FRONTMATTER + MARKDOWN BODY

SCHEMA

```
---
scope:          team                        # org | team
kind:           deploy-notes                # free-form category
title:          NMB Bird Street pilot deploy
created_at:     2026-11-05T14:32:03Z
updated_at:     2026-11-05T16:18:22Z
content_hash:   sha256:9e3d1a7f8e2b4c1d...  # hash of body (this version)
previous_hash:   sha256:f4a2b8c1...           # hash of previous body
attributed_to:  agent:glm-4.6               # who wrote this version
precondition:   sha256:f4a2b8c1...            # must match current content_hash
reversible:    true                         # can be reverted via git
---

# NMB Bird Street pilot deploy

## 2026-11-05 14:32 — initial deploy
- Edge node online at Bird Street / Cuyler Street site
- PLC tap N02 reading flow_in=41.3 L/s, flow_out=33.3 L/s
- 8.0 L/s missing → LEAK_ACTIVE receipt emitted
- Chain head: 9e3d1a7f8e2b4c1d...
```

WRITE PROTOCOL · OPTIMISTIC CONCURRENCY

5 STEPS

01	Read current file · compute currentHash = SHA256(body)
02	Check precondition · if stale → re-read, re-derive, retry
03	Compute new body · newHash = SHA256(new_body)
04	Write frontmatter · previous_hash=currentHash, attributed_to=self
05	Git commit · capture diff + agent name in author

PORTABILITY TEST

PASSED

```
$ tar -czf searm-memory.tar.gz MEMORY.md org/ team/ $ scp searm-memory.tar.gz new-machine:~/ new-machine$ tar -xzf searm-memory.tar.gz ✓ Memory fully portable · no DB · no SQL dump · just files
```

THREE TEMPORAL WINDOWS

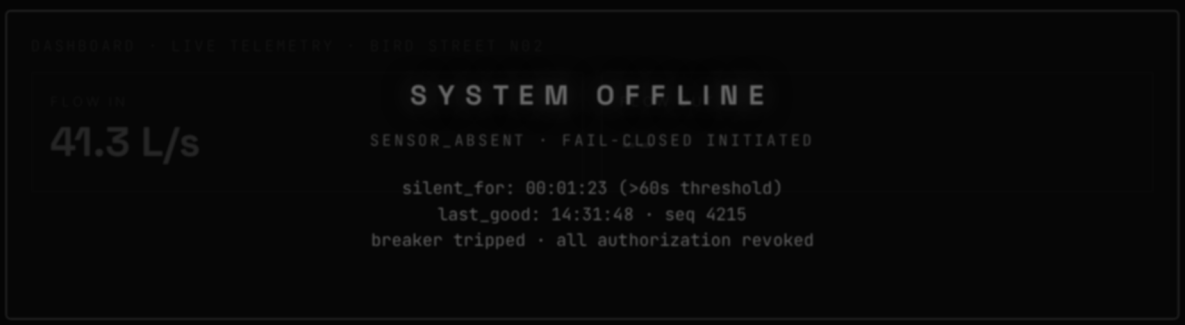
SKILL 07

WINDOW	DURATION	STATE	RECEIPT	UI
Live	<15s since last reading	LIVE	normal telemetry	value + timestamp
Stale	15s — 60s	SENSOR_ABSENT	absence receipt	SENSOR_ABSENT badge
Fail-closed	>60s	FAIL_CLOSED	heartbeat + breaker trip	red overlay + grayscale

ACTIVE FAILURE RECEIPTS · PER VERTICAL

5 VERTICALS

VERTICAL	RECEIPT	TRIGGER
Health	SENSOR_ABSENT	Sensor removed (HR=0 or no BLE >15s)
Finance	FRAUD_PATTERN_DETECTED	Beneficiary created <60s before high-value transfer
Industrial	LEAK_ACTIVE	flow_in - flow_out > threshold for >5s
Industrial	BREAKER_TRIPPED	Overcurrent or breaker state = TRIPPED
Infrastructure	NODE_UNREACHABLE	Server / DNS / edge node ping fails >15s
Community/Gov	WATER_LOSS_RECEIPT	Bulk flow_in - billed_out > threshold / 24h



[engine] 14:31:48 **TELEMETRY seq=4215 flow_in=41.3 flow_out=33.3** [engine] 14:32:03 no reading for 15s → emit SENSOR_ABSENT receipt (seq 4216) [engine] 14:32:48 no reading for 60s → emit HEARTBEAT (seq 4217) [engine] 14:32:48 **FAIL_CLOSED · breaker tripped · auth revoked** [engine] 14:33:11 **reading arrived but recovery requires reverification + operator ack**

THE FIVE VERTICALS · SAME ENGINE, DIFFERENT SKINS

SKILL 08

VERTICAL	SKIN (UI / DATA SOURCE)	ESTATE (WHAT IS PROVEN)	FAIL-CLOSED TRIGGER
Health / Wearables	Universal Avatar (Biometrics + Health Apps)	Human Vitals — HR, SpO2, HRV	SENSOR_ABSENT (0 BPM rejected)
Finance / Fraud	Payment Gateways & Core Banking	Transaction Sequences	FRAUD_PATTERN_DETECTED
Industrial / IoT	Digital Twins (Heavy Machinery)	Plant telemetry (flow, breaker, temp)	LEAK_ACTIVE / BREAKER_TRIPPED
Infrastructure	Uptime Dashboards (Hosting, DNS)	Server health, routing, uptime	NODE_UNREACHABLE
Community / Gov	Smart City Dashboards	Bulk utilities in vs billed out	WATER_LOSS_RECEIPT (8.0 L/s)

RECEIPT TYPES · CANONICAL STRINGS (CHAIN ENTRY TYPE FIELD)

12 TYPES

CORE TYPES

CANONICAL

type	VITALS	LIVE
type	TELEMETRY	LIVE
type	TRANSACTION	LIVE
type	UPTIME	LIVE
type	ABSENCE_RECEIPT	ABSENT
type	HEARTBEAT	LIVENESS

VERTICAL-SPECIFIC

CANONICAL

type	LEAK_RECEIPT	FAIL
type	BREAKER_RECEIPT	FAIL
type	FRAUD_RECEIPT	FAIL
type	WATER_LOSS_RECEIPT	FAIL
type	NODE_UNREACHABLE_RECEIPT	FAIL
type	TSA_ANCHOR	ANCHOR

MODE TAGS · EVERY VALUE CARRIES ONE

3 MODES

LIVE	Measured from a real sensor in production	LIVE
SIM	Deterministic simulation (meta.sim = true)	SIM
DEMO	Produced for demonstration (meta.demo = true)	DEMO

THE ANTI-INVENTION RULE

ENFORCED

FORBIDDEN — INVENTED COMPONENT IDS

✗

bad	sensor_42 · ENGINE_001 · motor_v2 · powertrain_a
good	plc_tap_N02 · ENGINE · Chassis #VW-8821 · flow_in
rule	If not in vocabulary → ask user to define before rendering

HARD BOUNDARY · READ-ONLY FUNCTION CODES

SKILL 09

MODBUS · ACCEPTABLE ✓

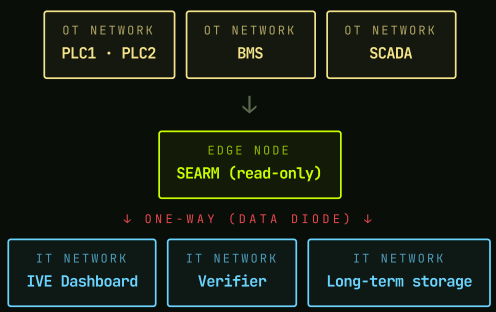
0x03	Read Holding Registers
0x04	Read Input Registers
0x01	Read Coils
0x02	Read Discrete Inputs

MODBUS · FORBIDDEN ✗

0x05	Write Single Coil
0x06	Write Single Register
0x0F	Write Multiple Coils
0x10	Write Multiple Registers

OT/IT NETWORK TOPOLOGY · DATA DIODE

TOPOLOGY



PILOT ACTIVATION CHECKLIST · ALL SIGNED BEFORE ACTIVATION

6/6

01	Client safety review — SIS unaffected	SIGNED
02	IT/OT security review — no new IT→OT paths	SIGNED
03	Change management logged with rollback plan	SIGNED
04	Read-only verification — function codes reviewed	SIGNED
05	Pilot agreement signed (read-only clause explicit)	SIGNED
06	Operator briefing — how to disconnect if concerned	SIGNED

```
$ grep -rE 'writeSingleRegister|writeMultipleRegisters' src/ (no matches found) ✓ $ grep -rE 'session\.write\(' src/ (no matches found) ✓ $ node edge.js --verify-readonly OPC-UA session: readOnly=true ✓ Subscribed nodes: 142 (all CurrentRead) ✓ Write calls: 0 ✓ ✓ EDGE NODE IS STRICTLY READ-ONLY
```

THE FOUR TAGS · EVERY CLAIM CARRIES ONE

SKILL 10

TAG	MEANING	EXAMPLE
[MEASURED]	Real, live sensor in production	flow_in=41.3 L/s [MEASURED: plc_tap_N02, 14:32:03Z, seq 4217]
[PUBLIC: source]	Public fact, attributable source	water loss 60.28% [PUBLIC: NMB Annual Report 2025/26, p47]
[DEMO]	Synthetic, for demonstration	HR 72 BPM [DEMO] · BP 129/74 [DEMO]
[VERIFY]	Asserted but not yet verified	Cubot BLE UUID 0x180F [VERIFY — confirm via nRF Connect]

THE "DEPLOYED FOR" RULE · WITHOUT SIGNED AGREEMENT

FORBIDDEN

FORBIDDEN PHRASES ✗

bad	"deployed for Nelson Mandela Bay"
bad	"deployed for Stitch"
bad	"deployed for Capitec"
bad	"in production at Aspen"
bad	"live at Fresenius"

ACCEPTABLE ALTERNATIVES ✓

good	"pilot in discussion with NMB"
good	"pilot in discussion with Stitch"
good	"pilot in discussion with Capitec"
good	"pilot in discussion with Aspen"
good	"pilot in discussion with Fresenius"

THREE-LAYER TAGGING · DEMO VALUES IN 3 PLACES

3 LAYERS

value	{ "heart_rate": 72, "sim": true, "demo_tag": "DEMO" }	DEMO
ledger	{ "type": "VITALS_SIM", "meta": { "demo": true } }	SIM
UI	visible DEMO badge in corner of panel	DEMO
untagged	FORBIDDEN — treated as forgery	✗

DRIFT AUDIT · PER RELEASE

PRE-SHIP

public README	all numbers tagged [PUBLIC: source] or removed	PASS
private pitch	all numbers tagged [MEASURED]/[PUBLIC]/[DEMO]/[VERIFY]	PASS
chain entries	all meta.mode = LIVE / SIM / DEMO	PASS
customer comms	no "deployed for" without signed agreement	PASS

CHAIN ENTRY · SALTED HASHES ONLY

SKILL 11

FORBIDDEN · RAW PHI ✗

```
{
  "type": "VITALS",
  "patient_name": "John Doe",
  "patient_id": "NMB-001-2026",
  "metrics": { "heart_rate": 72 }
}
// ✗ patient_name + patient_id = PHI
// ✗ cannot appear in chain
```

REQUIRED · SALTED HASH ✓

```
{
  "type": "VITALS",
  "subject_hash": "sha256(a3f8...||John Doe)",
  "metrics": { "heart_rate": 72 },
  "source": "wearable_device",
  "consent_receipt_seq": 4210,
  "meta": { "mode": "LIVE" }
}
// ✓ subject_hash breaks link to identity
// ✓ heart_rate is a measurement, not PHI
```

CONSENT RECEIPT · EVERY DATA COLLECTION HAS ONE

CONSENT

```
{
  "type": "CONSENT_RECEIPT",
  "subject_hash": "sha256(salt_subject_001 || subject_id)",
  "scope": "vitals_collection",
  "purpose": "operational_monitoring",
  "duration_days": 365,
  "withdrawable": true,
  "consent_method": "explicit_tap",
  "consented_at": "2026-11-05T14:31:00Z",
  "expires_at": "2027-11-05T14:31:00Z",
  "meta": { "signature": "...", "mode": "LIVE" }
}
```

CRYPTOGRAPHIC ERASURE · RIGHT TO BE FORGOTTEN

ERASURE

01	Subject requests erasure
02	Emit CRYPTOGRAPHIC_ERASURE receipt (names subject_hash + entries affected)
03	Destroy the salt in consent registry
04	**Chain entries remain · but subject_hash is now unlinkable**
result	Chain still verifies cryptographically · link to subject broken

GXP BOUNDARY · INDEPENDENT EVIDENCE COPY

REGULATED

01	SEARM does not connect to validated system (passive taps)
02	SEARM evidence not used for GxP decisions
03	SEARM does not feed data back into validated system
04	SEARM sits beside — never inside — the GxP boundary
clause	**Pilot agreement explicitly states "independent evidence copy"**

THE ONE QUESTION

SKILL 12

> Does this change **what is physically true**, or only **how it is shown**?
>
> If "what is physically true" → FORBIDDEN · If "how it is shown" → PERMITTED

FIVE-QUESTION DRIFT CHECK

5 GATES

01	Does it change what is physically true? (interpolate, fabricate, invent ID, modify chain)	NO
02	Does it require phone-home, API key, or login to verify?	NO
03	Does it break the chain's cryptographic integrity? (canonical, hash, sig, domain)	NO
04	Does it touch client OT, store raw PHI, or claim "deployed" without signed agreement?	NO
05	Does the change still hand the user the receipt and the verifier?	YES

✓ SHIP IT

ALL 5 GATES PASSED · NO DRIFT DETECTED

THE MASTER GATE · INVOKES ALL 11 OTHER SKILLS

11 REFS

Q1	→ Skill 01 (No-Primitives Law)
Q2	→ Skill 02 (Local-First) · Q5 (verifier runs on every surface)
Q3	→ Skill 03 (WORM Chain) · Q5 (chain still verifies)
Q4	→ Skill 04 (Cross-Platform) · 05 (Licensing) · 08 (Vocab) · 09 (Edge) · 10 (Claims) · 11 (POPIA)
Q5	→ Skill 06 (Memory) · 07 (Fail-Closed)

```
$ ./scripts/drift-check.sh --since v1.2.0
Walking 47 commits since last release...
Each commit checked against the 5-question gate...
✓ 47/47 commits passed drift check
✓ Release v1.3.0 cleared to ship
Drift-check results recorded in release notes.
```

HDRP VOLUME STACK · THE HOUSE LOOK

SKILL 13

POST-PROCESSING — GLOBAL VOLUME

tonemapping	ACES
white_balance	6500K outdoor · 4500K indoor
saturation	-5 (slightly muted)
contrast	+8 (punchier)
bloom	threshold 1.0 · intensity 0.2
DOF	OFF (technical twins)
motion_blur	OFF (looks like photograph)
SSR	ON (or Ray Traced on RT GPU)
GTAO	ON · intensity 0.5
contact_shadows	ON · length 0.15
volumetric_fog	ON (outdoor atmosphere)
anti-aliasing	TAA (temporal)



LIGHTING RECIPE

RECIPE

sky	HDRI Sky (real on-site capture, 8K equirect EXR)
sun	Directional Light matched to HDRI sun position
indirect	APV (Adaptive Probe Volumes) — replaces baked probes
interior	Area Lights (Rect, Tube, Disc) — realistic specular
emissive	HDR intensity > 1.0 for screens, LEDs, glow signs

PERFORMANCE BUDGETS · PER DEVICE

6 TARGETS

DEVICE	FPS	RESOLUTION	RT
Desktop (RTX 3070+)	60-120	1440p-4K	Yes
Desktop (GTX 1060+)	60	1080p	No
Apple Vision Pro	90	4K/eye	Metal RT
Meta Quest 3	72	2064×2208/eye	No
iPhone 15 Pro+	60	1080p	Metal RT
Android (SD 8 Gen 3+)	60	1080p	Vulkan RT

THE 10-STAGE PIPELINE

SKILL 14

01	Concept / scan / reference (real-world photos + ColorChecker)
02	Blockout — low-poly silhouette at real-world scale (1:1 meters)
03	High-poly sculpt — ZBrush / Blender sculpt (5M-15M tris hero)
04	Retopology — clean low-poly mesh (all-quads on hero)
05	UV unwrap — single atlas, 80% utilisation, no overlaps
06	Bake — Normal, AO, Curvature, Position, Bent Normal (4K)
07	PBR material — Metallic-Roughness (Substance Painter)
08	Decimation + LODs — LOD0/1/2/3 with invisible transitions
09	Export GLB 2.0 — embedded textures, Meshopt compression
10	Publish to Sketchfab — CC-BY-4.0, author attribution

TRIANGLE TARGETS · PER ASSET CLASS

4 CLASSES

ASSET CLASS	LOD0	LOD1	LOD2	TEXTURES
Hero (vehicle, character)	30k-80k	10k-25k	3k-8k	4K
Environment (building, machine)	5k-30k	2k-8k	500-2k	2K
Prop (chair, tool)	500-5k	200-1k	50-200	1K
Background (distant)	100-500	50-100	10-50	512px

GLB 2.0 EXPORT · GLTF-VALIDATOR PASSES

VALIDATED

$ gltf-validator isuzu-dmax.glb **✓ Valid GLB 2.0 ✓ Materials use metallic-roughness workflow ✓ No unused materials or textures ✓ No missing UVs on materials that need them ✓ No overlapping UVs (within tolerance)** ⚠ Hint: texture 4096² but model viewed at 2m — consider 2048² $ ls -lh isuzu-dmax.glb -rw-r--r-- 28M isuzu-dmax.glb (Meshopt compressed)

SKETCHFAB PUBLISH · ATTRIBUTION CHAIN

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title	"Isuzu D-Max"
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BIRD STREET SITE · PHOTOREAL MAP

SKILL 15



THE STACK

STACK

RENDERER

CANONICAL

map	**mapbox-gl / maplibre-gl**
base	Satellite Streets
terrain	Terrain-DEM v1 (raster)
buildings	fill-extrusion (vector tiles)
twin	three.js + threebox (GLB at lat/lng)
atmosphere	Mapbox Fog + custom sky
camera	FreeCamera (3D position + lookAt)

STYLE · DARK INDUSTRIAL

SEARM

satellite	saturation -0.1 · contrast +0.05
roads	**#c8ff00 · opacity 0.4**
buildings_3d	#1a1a1a · opacity 0.6
light	ochre #f3e38a · intensity 0.6
sky	#0a0e09 → #1a1a1a horizon
fog	range [1,12] · color #050505
stars	0.0 (urban) · 0.5 (rural)

OFFLINE TILE CACHING · LOCAL-FIRST

OFFLINE

```
$ offlineManager.downloadRegion({ name: 'nmb-bird-street', styleURL: 'mapbox://styles/mapbox/satellite-streets-v12', bounds: [[25.55, -33.99], [25.70, -33.91]], minZoom: 12, maxZoom: 18 }) ✓ Region downloaded (47 MB cached locally) → Field deployment works without internet
```

THE SIGNAL STACK · REAL-WORLD INPUTS

SKILL 16

SIGNAL	SOURCE	ACCURACY	USE
GPS	navigator.geolocation	5-10m (phone)	Anchor twin to real location
Compass	DeviceOrientationEvent	1-2° (calibrated)	Camera bearing matches facing direction
Ambient light	AmbientLightSensor (W3C)	±10 lux	Match twin exposure to room
IMU	accelerometer + gyroscope	1° (Kalman filtered)	Phone orientation drives twin camera
Time of day	Date.now()	<1ms	Sun position via SunCalc
Weather	Open-Meteo API (keyless)	1km grid · 5-min	Rain → wet surfaces · fog → volumetrics
Depth camera	ARKit / ARCore / WebXR	mm-level (LiDAR)	Real objects occlude virtual twin
World tracking	ARKit / ARCore	mm-level	Anchor twin to real ground

TIME-OF-DAY SUN MATCHING · SUNCALC

SUN

```
$ node -e "const s = SunCalc.getPosition(new Date(), -33.96, 25.61); console.log(s);" { azimuth: 1.83, altitude: 0.91 } azimuth 1.83 rad = 104.8° (east-southeast) altitude 0.91 rad = 52.1° (high sun) $ node -e "console.log(sunColor(altitude_deg));" → altitude 52° > 15° → daylight white #ffffff 06:00 → 5° → golden hour #ffd700 18:00 → 0° → sunset #ff8c42 19:00 → -3° → twilight #ff6600 22:00 → -30° → night #0a0e2a (moonlight)
```

AR MODE · TWIN ANCHORED TO REAL GROUND

AR

trigger	User within 100m of twin's GPS coordinates
session	navigator.xr.requestSession('immersive-ar')
placement	Hit-test against floor → anchor twin at pose
occlusion	**Depth API → real objects hide virtual twin**
lighting	**Match sun position from SunCalc + ambient lux sensor**
interaction	User physically walks around the virtual twin

WEATHER INTEGRATION · OPEN-METEO (KEYLESS)

WEATHER

```
$ curl 'https://api.open-meteo.com/v1/forecast?latitude=-33.96&longitude=25.61&current=temperature_2m,precipitation,cloud_cover' { "current": { "temperature_2m": 22.4, "precipitation": 0.0, "cloud_cover": 35 } } $ applyWeather(twin, weather) → no rain · surfaces stay dry (roughness unchanged) → cloud cover 35% · sun intensity = 0.755
```

BUILD MATRIX · ONE CODEBASE → THREE NATIVE BINARIES

SKILL 17

⌥

TAURI · DESKTOP

14 MB

.DMG · .MSI · .APPIMAGE

⌘

CAPACITOR · iOS

78 MB

.IPA · METAL · HEALTHKIT

●

CAPACITOR · ANDROID

85 MB

.AAB · VULKAN · BLE

WHY TAURI OVER ELECTRON

RUST

TAURI	~10MB
binary	14 MB
memory	82 MB idle
webview	System (no bundled Chromium)
backend	Rust (memory safe)
updates	With OS webview

ELECTRON	~150MB
binary	150 MB
memory	200 MB idle
webview	Bundled Chromium
backend	Node.js (GC pauses)
updates	Manual Chromium re-package

WEBGL CONTEXT PRESERVATION · BACKGROUND → FOREGROUND

ROBUST

```
const gl = canvas.getContext('webgl2', {
  antialias: true,
  preserveDrawingBuffer: true,    // for screenshots
  powerPreference: 'high-performance',
  failIfMajorPerformanceCaveat: false,  // software fallback
  desynchronized: true               // reduce latency
});

canvas.addEventListener('webglcontextlost', (e) => {
  e.preventDefault();
  renderer.setAnimationLoop(null);  // pause
});

canvas.addEventListener('webglcontextrestored', () => {
  initRenderer();            // re-init state
  renderer.setAnimationLoop(animate);  // resume
});
```

BUNDLE SIZE REDUCTION · KTX2 + MESHOPT

70% SMALLER

```
$ toktx --bcmp --uastc 4 isuzu-dmax_body.ktx2 isuzu-dmax_body.png PNG 4096²: 12.4 MB KTX2/Basis: 3.5 MB (-72%) $ gltf-transform optimize isuzu-dmax.glb isuzu-dmax-opt.glb --compress meshopt GLB raw: 28 MB GLB Meshopt: 11 MB (-61%)
```

THE PBR MAP STACK · METALLIC-ROUGHNESS WORKFLOW

SKILL 18

MAP	CHANNELS	RANGE	PURPOSE
Albedo	RGB	0-1	Raw surface color · no baked lighting
Normal	RGB (tangent)	0-1 → -1..1	High-frequency surface detail
Roughness	Grayscale	0 (mirror) - 1 (diffuse)	Specular blur
Metallic	Grayscale	0 or 1	Metal vs dielectric
AO	Grayscale	0-1	Contact shadows · multiplied in shader
Emissive	RGB (HDR)	0 to >1	Glow · intensity >1 for bloom
Height	Grayscale	0-1	Parallax / displacement (optional)

MATERIAL SWATCHES · COMMON RECIPES

3 SWATCHES

GALVANISED STEEL

albedo	#b8b8b8
roughness	0.4-0.6
metallic	1.0
normal	subtle bumps
variation	±5% roughness

CAR PAINT · LIME

albedo	#c8ff00
roughness	0.15
metallic	0.0
clearcoat	1.0 (0.05)
normal	orange peel

WET ASPHALT

albedo	#1a1a1a
roughness	0.3 (wet)
metallic	0.0
puddles	roughness 0.05
variation	dry 0.7

CAPTURE METHODS · THREE SOURCES

3 METHODS

PHOTOGRAMMETRY — HERO

capture	Polarised light (cross + co)
derive	albedo + roughness + normal
tool	Substance 3D Sampler
use	**Hero assets**

PROCEDURAL — TILED

author	Substance Designer node graph
output	Tileable · resolution-independent
params	rust amount, scratch density
use	**Concrete, asphalt, large areas**

HAND-PAINTED — STORY

author	Substance Painter
base	Baked maps (from Skill 14)
generators	Dirt · Edge Wear · Grunge
use	**Hero storytelling**

VALIDATION · TEST SCENE WITH NEUTRAL HDRI

VALIDATED

```
$ node validate-material.js isuzu-dmax_body/ Loading RoomEnvironment (neutral HDRI)... Applying material to test sphere... Rendering from 4 angles... ✓ Albedo: no baked lighting detected ✓ Roughness: variation present (0.15 - 0.45) ✓ Metallic: binary (0 or 1, anti-aliased edges) ✓ Normal: OpenGL convention (+Y up) ✓ AO: separate map, not baked into albedo ✓ Material validates · ready for GLB export
```